* Revision:

- data analysis / data visualization
- missing values
- outliers

* Today's class:

① Two fancy words:

a) Univariate analysis : Analysis single variable.

b) Multivariate analysis : Analyzing multiple variables together.

② Categorical Encoding techniques:

- What is Encoding ?

- Why Encoding is Require ?

@ One-hot Encoding:

- creates a seperate binary column for each category.

Ex:

color

↳ three cate: Red, Blue, Green (Single column) → convert it into three column.

color → Red Blue Green

- Data Example

color	Red	blue	Green
Red	1	0	0
Red	1	0	0
blue	0	1	0
Green	0	0	1

Pros

- No artificial order between categories.
- Easy to understand and implement.
- Works well with linear models, SVM, and many other algorithms.

Cons

- Creates many columns when there are many unique categories.
- Can increase memory usage.
- Unseen categories need to be handled during inference.

Use when: The feature has a small or moderate number of nominal categories, such as gender, color, or payment method.

(b) Ordinal Encoding:

→ Assigns numbers according to a meaningful order.

Cate: low → 1
mid → 2
high → 3

IQ
low
mid
high

→

IQ
1
2
3

Pros

- Simple and memory-efficient.
- Preserves the intended category order.
- Useful for genuinely ordered categories.

Cons

- The numeric distances may be misleading: the difference between 1 and 2 isn't necessarily equal to the difference between 3 and 4.
- Can mislead linear models if the numeric effect isn't appropriate.

Use when: Categories have a genuine order, such as low/medium/high, satisfaction ratings, or education levels.

ⓒ Label Encoding :

- assigns an integer to each category.



Pros

- Simple and compact.
- Doesn't increase the number of columns.
- Can be useful with algorithms that explicitly support categorical features or for encoding target labels.

Cons

- Introduces an artificial numerical order for nominal categories.
- Can harm linear models, distance-based algorithms, and other models that interpret the values numerically.

Use when: Encoding the **target labels** in a classification problem, or when the model explicitly supports categorical values. For ordinary nominal input features, prefer one-hot encoding or another suitable method.

→ There are so many other techn.

② Feature Engineering :

- Feature Engineering is the process of creating, transforming or selecting input features to help a ML model learn useful pattern from data.

@ Creating New features :

Area (sq. ft.)	Bedrooms	Bathrooms	Price
1,000	2	1	₹30 lakh
1,500	3	2	₹45 lakh
2,000	4	3	₹60 lakh

(b) Date & time :

(c) Transformation

(d) feature scaling & Normalization

(e) Encoding categorical features.

(f) Feature selection

(g) Polynomial features

(h) Binning :

→ converts continuous variable into intervals.

Ex: Grading

Marks : 90 + : A

80 < m < 90 : B

70 < m < 80 : C

:

→ Feature Engi. is not only Applies to data we got.

Netflix ⟶ Review system before



Netflix: ⟶ New Review system

like dislike

Simple UI change : 200% more Rating activity.

③ Duplicate values:

→ Pending:

- Scaling & Norm
- Transform
- Feature selection

(PCA / Regul / model based / statistical test)

